ULTIMATE GEMMA EVOLUTION MATRIX (176 Configs)

config_label	agent	functional_correctness	branch_coverage	mutation_score	prompt_tokens	completion_tokens	total_tokens	avg_tokens_per_task
chatgpt-oss	baseline	0.56712	nan	1	6310	38588	44898	1795.92
chatgpt-oss	cot	0.81732	nan	0.89468	7610	30064	37674	1506.96
chatgpt-oss	few_shot	0.0116	nan	1	10660	46616	57276	2291.04
chatgpt-oss:zero_shot	baseline	0.56712	nan	1	6310	38588	44898	1795.92
chatgpt-oss:zero_shot	cot	0.81732	nan	0.89468	7610	30064	37674	1506.96
chatgpt-oss:zero_shot	few_shot	0.0116	nan	1	10660	46616	57276	2291.04
cot:chatgpt-oss/chatgpt-oss	swarm	0.8333	nan	0.4983	12467	16692	29159	2915.9
cot:gemma-27b/chatgpt-oss	swarm	1	nan	0.165	10649	2479	13128	1312.8
cot:gemma-27b/llama3-70b	actor_critic	0.4035	nan	1	33445	6648	40093	4009.3
cot:gemma-27b/llama3-70b	swarm	0.9	nan	0.1983	8837	540	9377	937.7
cot:llama3-8b	actor_critic	0.4365	nan	1	5894	5297	11191	5595.5
cot:llama3-8b	baseline	0.5975	nan	1	479	1216	1695	847.5
cot:llama3-8b	hybrid	0.3359	nan	1	27751	48304	76055	3802.75
cot:llama3-8b	swarm	0.5	nan	0.7	1991	768	2759	1379.5
gemma-12b:cot	actor_critic	0.64268	nan	0.888	71398	467	71865	2874.6
gemma-12b:cot	adversarial	0.67672	nan	0.896	22267	0	22267	890.68
gemma-12b:cot	atomic_swarm	0.75336	nan	0.76664	37910	0	37910	1516.4
gemma-12b:cot	baseline	0.63076	nan	0.884	28089	8192	36281	1451.24
gemma-12b:cot	coa	0.78476	nan	0.79668	46295	0	46295	1851.8
gemma-12b:cot	competitive	0	nan	1	0	0	0	0
gemma-12b:cot	consensus	0.19884	nan	0.936	92661	410	93071	3722.84
gemma-12b:cot	hybrid	0.61248	nan	0.968	78344	0	78344	3133.76
gemma-12b:cot	self_healing	0.23584	nan	0.96668	25708	2988	28696	1147.84
gemma-12b:cot	soa	0.29348	nan	0.968	41790	0	41790	1671.6
gemma-12b:cot	swarm	0.72176	nan	0.482	34570	0	34570	1382.8
gemma-12b:few_shot	actor_critic	0.57484	nan	0.92	57937	0	57937	2317.48
gemma-12b:few_shot	adversarial	0.20116	nan	0.93	49515	372	49887	1995.48
gemma-12b:few_shot	atomic_swarm	0.74004	nan	0.78664	31195	0	31195	1247.8
gemma-12b:few_shot	baseline	0.69032	nan	0.914	53977	0	53977	2159.08
gemma-12b:few_shot	coa	0.81384	nan	0.78336	30362	0	30362	1214.48
gemma-12b:few_shot	competitive	0	nan	1	0	0	0	0
gemma-12b:few_shot	consensus	0.17828	nan	0.936	67824	213	68037	2721.48

config_label	agent	functional_correctness	branch_coverage	mutation_score	prompt_tokens	completion_tokens	total_tokens	avg_tokens_per_task
gemma-12b:few_shot	hybrid	0.62008	nan	0.85668	66695	0	66695	2667.8
gemma-12b:few_shot	self_healing	0.27656	nan	0.93468	30316	15733	46049	1841.96
gemma-12b:few_shot	soa	0.31904	nan	0.968	25415	0	25415	1016.6
gemma-12b:few_shot	swarm	0.76	nan	0.38	37547	0	37547	1501.88
gemma-12b:scot	actor_critic	0.44632	nan	0.96	56312	729	57041	2281.64
gemma-12b:scot	adversarial	0.56664	nan	0.81	44317	0	44317	1772.68
gemma-12b:scot	atomic_swarm	0.74004	nan	0.78664	33747	0	33747	1349.88
gemma-12b:scot	baseline	0.69888	nan	0.78468	44263	0	44263	1770.52
gemma-12b:scot	coa	0.8132	nan	0.79668	34498	0	34498	1379.92
gemma-12b:scot	competitive	0	nan	1	0	0	0	0
gemma-12b:scot	consensus	0.15556	nan	0.936	83193	8683	91876	3675.04
gemma-12b:scot	hybrid	0.47484	nan	0.97	47234	0	47234	1889.36
gemma-12b:scot	self_healing	0.25832	nan	0.96668	27859	19785	47644	1905.76
gemma-12b:scot	soa	0.28152	nan	0.936	53004	0	53004	2120.16
gemma-12b:scot	swarm	0.74	nan	0.374	44373	0	44373	1774.92
gemma-12b:zero_shot	actor_critic	0.18088	nan	0.97	66798	0	66798	2671.92
gemma-12b:zero_shot	adversarial	0.29176	nan	0.97	34475	0	34475	1379
gemma-12b:zero_shot	atomic_swarm	0.76672	nan	0.76664	37805	0	37805	1512.2
gemma-12b:zero_shot	baseline	0.48372	nan	0.928	20913	0	20913	836.52
gemma-12b:zero_shot	coa	0.80352	nan	0.77268	27861	0	27861	1114.44
gemma-12b:zero_shot	competitive	0	nan	1	0	0	0	0
gemma-12b:zero_shot	consensus	0.20144	nan	0.944	74278	0	74278	2971.12
gemma-12b:zero_shot	hybrid	0.04	nan	1	66485	0	66485	2659.4
gemma-12b:zero_shot	self_healing	0.23416	nan	0.93468	32994	7304	40298	1611.92
gemma-12b:zero_shot	soa	0.28172	nan	0.968	37499	0	37499	1499.96
gemma-12b:zero_shot	swarm	0.84	nan	0.31	45745	0	45745	1829.8
gemma-1b:cot	actor_critic	0	nan	1	40864	18110	58974	2358.96
gemma-1b:cot	adversarial	0.01716	nan	1	23977	9816	33793	1351.72
gemma-1b:cot	atomic_swarm	0	nan	1	27790	22422	50212	2008.48
gemma-1b:cot	baseline	0	nan	1	26514	24352	50866	2034.64
gemma-1b:cot	coa	0	nan	1	73919	34758	108677	4347.08
gemma-1b:cot	competitive	0	nan	1	0	0	0	0
gemma-1b:cot	consensus	0.11428	nan	1	73301	32410	105711	4228.44
gemma-1b:cot	hybrid	0	nan	1	53279	18787	72066	2882.64

config_label	agent	functional_correctness	branch_coverage	mutation_score	prompt_tokens	completion_tokens	total_tokens	avg_tokens_per_task
gemma-1b:cot	self_healing	0.01716	nan	1	22184	8168	30352	1214.08
gemma-1b:cot	soa	0.06472	nan	1	57715	55247	112962	4518.48
gemma-1b:cot	swarm	0.09992	nan	0.96	67483	91516	158999	6359.96
gemma-1b:few_shot	actor_critic	0.48384	0	0.648	87109	45205	132314	5292.56
gemma-1b:few_shot	adversarial	0.41332	nan	0.706	21835	388	22223	888.92
gemma-1b:few_shot	atomic_swarm	0	nan	1	22788	14709	37497	1499.88
gemma-1b:few_shot	baseline	0.10932	nan	0.928	31138	44518	75656	3026.24
gemma-1b:few_shot	coa	0	nan	1	21202	15230	36432	1457.28
gemma-1b:few_shot	competitive	0	nan	1	0	0	0	0
gemma-1b:few_shot	consensus	0.17504	nan	0.96	74676	14391	89067	3562.68
gemma-1b:few_shot	hybrid	0.4312	nan	0.738	74039	46275	120314	4812.56
gemma-1b:few_shot	self_healing	0.02	nan	1	22134	294	22428	897.12
gemma-1b:few_shot	soa	0.05224	nan	1	26743	29071	55814	2232.56
gemma-1b:few_shot	swarm	0.15428	nan	0.92	32245	29752	61997	2479.88
gemma-1b:scot	actor_critic	0.15272	nan	1	85569	66530	152099	6083.96
gemma-1b:scot	adversarial	0.2194	nan	1	26608	1491	28099	1123.96
gemma-1b:scot	atomic_swarm	0	nan	1	58896	28032	86928	3477.12
gemma-1b:scot	baseline	0.05668	nan	1	24621	25111	49732	1989.28
gemma-1b:scot	coa	0	nan	1	40254	42752	83006	3320.24
gemma-1b:scot	competitive	0	nan	1	0	0	0	0
gemma-1b:scot	consensus	0.11736	nan	1	78729	29925	108654	4346.16
gemma-1b:scot	hybrid	0.26904	nan	1	43925	35351	79276	3171.04
gemma-1b:scot	self_healing	0.01716	nan	1	19804	1564	21368	854.72
gemma-1b:scot	soa	0.07004	nan	1	38264	40466	78730	3149.2
gemma-1b:scot	swarm	0.07076	nan	1	43690	33069	76759	3070.36
gemma-1b:zero_shot	actor_critic	0.2594	nan	1	70646	27553	98199	3927.96
gemma-1b:zero_shot	adversarial	0.25384	nan	1	33729	13814	47543	1901.72
gemma-1b:zero_shot	atomic_swarm	0.04	nan	0.96	40442	20924	61366	2454.64
gemma-1b:zero_shot	baseline	0.08636	nan	1	37400	16780	54180	2167.2
gemma-1b:zero_shot	coa	0	nan	1	34831	21214	56045	2241.8
gemma-1b:zero_shot	competitive	0	nan	1	0	0	0	0
gemma-1b:zero_shot	consensus	0.15336	nan	1	69772	28139	97911	3916.44

config_label	agent	functional_correctness	branch_coverage	mutation_score	prompt_tokens	completion_tokens	total_tokens	avg_tokens_per_task
gemma-1b:zero_shot	hybrid	0.29944	nan	0.96	93150	45607	138757	5550.28
gemma-1b:zero_shot	self_healing	0.03048	nan	1	25043	11748	36791	1471.64
gemma-1b:zero_shot	soa	0.05672	nan	1	39034	30620	69654	2786.16
gemma-1b:zero_shot	swarm	0.032	nan	1	49403	38951	88354	3534.16
gemma-27b:cot	actor_critic	0.65072	nan	0.97	60254	1077	61331	2453.24
gemma-27b:cot	adversarial	0.619292	nan	0.96875	40120	10144	50264	2094.33
gemma-27b:cot	atomic_swarm	0.82676	nan	0.54804	46407	0	46407	1856.28
gemma-27b:cot	baseline	0.7766	nan	0.72068	32701	0	32701	1308.04
gemma-27b:cot	coa	0.71652	0	0.738	46572	0	46572	1862.88
gemma-27b:cot	competitive	0	nan	1	0	0	0	0
gemma-27b:cot	consensus	0.72448	nan	0.87332	84173	12344	96517	3860.68
gemma-27b:cot	hybrid	0.65664	nan	0.938	67864	0	67864	2714.56
gemma-27b:cot	self_healing	0.06332	nan	0.96	58222	33185	91407	3656.28
gemma-27b:cot	soa	0.76096	nan	0.858	28498	0	28498	1139.92
gemma-27b:cot	swarm	0.75084	nan	0.49932	62875	299	63174	2526.96
gemma-27b:few_shot	actor_critic	0.76532	nan	0.85	58102	10732	68834	2753.36
gemma-27b:few_shot	adversarial	0.76376	nan	0.73668	39576	25729	65305	2612.2
gemma-27b:few_shot	atomic_swarm	0.83676	nan	0.54804	52143	11759	63902	2556.08
gemma-27b:few_shot	baseline	0.77332	nan	0.754	46421	0	46421	1856.84
gemma-27b:few_shot	coa	0.73376	0	0.634	48935	7640	56575	2263
gemma-27b:few_shot	competitive	0	nan	1	0	0	0	0
gemma-27b:few_shot	consensus	0.72848	nan	0.91332	88551	18586	107137	4285.48
gemma-27b:few_shot	hybrid	0.76796	nan	0.93668	87723	17264	104987	4199.48
gemma-27b:few_shot	self_healing	0.10856	nan	0.928	50284	4053	54337	2173.48
gemma-27b:few_shot	soa	0.75332	nan	0.858	43457	2909	46366	1854.64
gemma-27b:few_shot	swarm	0.8	nan	0.34064	44502	6715	51217	2048.68
gemma-27b:scot	actor_critic	0.72008	0	0.738	49429	12613	62042	2481.68
gemma-27b:scot	adversarial	0.77404	nan	0.746	20841	9768	30609	1224.36
gemma-27b:scot	atomic_swarm	0.82676	nan	0.54804	66319	492	66811	2672.44
gemma-27b:scot	baseline	0.80712	nan	0.66468	63714	0	63714	2548.56
gemma-27b:scot	coa	0.72576	0	0.738	52783	565	53348	2133.92
gemma-27b:scot	competitive	0	nan	1	0	0	0	0
gemma-27b:scot	consensus	0.6868	nan	0.91332	70815	15499	86314	3452.56
gemma-27b:scot	hybrid	0.80068	nan	0.74468	58924	16169	75093	3003.72

config_label	agent	functional_correctness	branch_coverage	mutation_score	prompt_tokens	completion_tokens	total_tokens	avg_tokens_per_task
gemma-27b:scot	self_healing	0.10856	nan	0.92668	32274	21396	53670	2146.8
gemma-27b:scot	soa	0.77072	nan	0.858	40959	612	41571	1662.84
gemma-27b:scot	swarm	0.76644	nan	0.45868	70914	2304	73218	2928.72
gemma-27b:zero_shot	actor_critic	0.67484	nan	0.89	70947	606	71553	2862.12
gemma-27b:zero_shot	adversarial	0.69348	nan	0.85668	37584	2477	40061	1602.44
gemma-27b:zero_shot	atomic_swarm	0.84008	nan	0.52404	29584	0	29584	1183.36
gemma-27b:zero_shot	baseline	0.68356	nan	0.85	38686	0	38686	1547.44
gemma-27b:zero_shot	coa	0.72724	0	0.674	31760	0	31760	1270.4
gemma-27b:zero_shot	competitive	0	nan	1	0	0	0	0
gemma-27b:zero_shot	consensus	0.6848	nan	0.91332	84879	1183	86062	3442.48
gemma-27b:zero_shot	hybrid	0.6942	nan	0.89	96072	0	96072	3842.88
gemma-27b:zero_shot	self_healing	0.0646	nan	0.96	38658	15560	54218	2168.72
gemma-27b:zero_shot	soa	0.7762	nan	0.858	26703	0	26703	1068.12
gemma-27b:zero_shot	swarm	0.72	nan	0.434	32650	0	32650	1306
gemma-4b:cot	actor_critic	0.5588	nan	0.92	36214	9274	45488	1819.52
gemma-4b:cot	adversarial	0.62064	nan	0.92	39557	126	39683	1587.32
gemma-4b:cot	atomic_swarm	0.78004	nan	0.582	33919	0	33919	1356.76
gemma-4b:cot	baseline	0.63156	nan	0.85	24291	1240	25531	1021.24
gemma-4b:cot	coa	0.80808	nan	0.766	29605	372	29977	1199.08
gemma-4b:cot	competitive	0	nan	1	0	0	0	0
gemma-4b:cot	consensus	0.57472	nan	1	85090	1446	86536	3461.44
gemma-4b:cot	hybrid	0.53876	nan	0.92	57980	413	58393	2335.72
gemma-4b:cot	self_healing	0.61108	nan	0.92	41339	0	41339	1653.56
gemma-4b:cot	soa	0.62712	nan	0.97	33680	975	34655	1386.2
gemma-4b:cot	swarm	0.67024	nan	0.77	51568	294	51862	2074.48
gemma-4b:few_shot	actor_critic	0.79108	nan	0.816	45060	8192	53252	2130.08
gemma-4b:few_shot	adversarial	0.80528	nan	0.888	39485	0	39485	1579.4
gemma-4b:few_shot	atomic_swarm	0.75336	nan	0.574	31912	0	31912	1276.48
gemma-4b:few_shot	baseline	0.78872	nan	0.858	37693	2813	40506	1620.24
gemma-4b:few_shot	coa	0.7896	nan	0.766	26860	8842	35702	1428.08
gemma-4b:few_shot	competitive	0	nan	1	0	0	0	0
gemma-4b:few_shot	consensus	0.61332	nan	0.89	85605	0	85605	3424.2

config_label	agent	functional_correctness	branch_coverage	mutation_score	prompt_tokens	completion_tokens	total_tokens	avg_tokens_per_task
gemma-4b:few_shot	hybrid	0.70764	nan	0.92	82149	8192	90341	3613.64
gemma-4b:few_shot	self_healing	0.62184	nan	0.92	33592	1376	34968	1398.72
gemma-4b:few_shot	soa	0.54	nan	1	30096	0	30096	1203.84
gemma-4b:few_shot	swarm	0.86	nan	0.244	20896	0	20896	835.84
gemma-4b:scot	actor_critic	0.59744	nan	0.93	55867	0	55867	2234.68
gemma-4b:scot	adversarial	0.38424	nan	0.96	41152	0	41152	1646.08
gemma-4b:scot	atomic_swarm	0.76716	nan	0.57732	25814	0	25814	1032.56
gemma-4b:scot	baseline	0.626	nan	0.82468	28522	0	28522	1140.88
gemma-4b:scot	coa	0.80632	nan	0.756	38039	582	38621	1544.84
gemma-4b:scot	competitive	0	nan	1	0	0	0	0
gemma-4b:scot	consensus	0.56888	nan	0.93	86328	8192	94520	3780.8
gemma-4b:scot	hybrid	0.62352	nan	0.85668	59020	168	59188	2367.52
gemma-4b:scot	self_healing	0.58668	nan	0.92	28576	357	28933	1157.32
gemma-4b:scot	soa	0.59964	nan	1	23632	1169	24801	992.04
gemma-4b:scot	swarm	0.66884	nan	0.73	59007	0	59007	2360.28
gemma-4b:zero_shot	actor_critic	0.58208	nan	0.97	60405	17704	78109	3124.36
gemma-4b:zero_shot	adversarial	0.61032	nan	0.81	39657	8971	48628	1945.12
gemma-4b:zero_shot	atomic_swarm	0.78052	nan	0.564	38439	1109	39548	1581.92
gemma-4b:zero_shot	baseline	0.64592	nan	0.85	33270	29922	63192	2527.68
gemma-4b:zero_shot	coa	0.78872	nan	0.766	36873	12458	49331	1973.24
gemma-4b:zero_shot	competitive	0	nan	1	0	0	0	0
gemma-4b:zero_shot	consensus	0.61848	nan	0.85	79058	294	79352	3174.08
gemma-4b:zero_shot	hybrid	0.49	nan	0.93	72638	12744	85382	3415.28
gemma-4b:zero_shot	self_healing	0.61376	nan	0.92	26007	0	26007	1040.28
gemma-4b:zero_shot	soa	0.58884	nan	0.96	18683	9464	28147	1125.88
gemma-4b:zero_shot	swarm	0.72644	nan	0.738	41794	9301	51095	2043.8
llama3-3b	swarm	0.6666	nan	0.9	4029	1624	5653	1130.6
llama3-3b:zero_shot	swarm	0.6666	nan	0.9	4029	1624	5653	1130.6
llama3-70b	baseline	0.85055	nan	0.7175	4336	6474	10810	540.5
llama3-70b:zero_shot	baseline	0.85055	nan	0.7175	4336	6474	10810	540.5
llama3_3b	atomic_swarm	0.3335	nan	1	0	0	0	0
llama3_3b	baseline	0.8005	nan	1	0	0	0	0
llama3_3b:zero_shot	atomic_swarm	0.3335	nan	1	0	0	0	0

config_label	agent	functional_correctness	branch_coverage	mutation_score	prompt_tokens	completion_tokens	total_tokens	avg_tokens_per_task
llama3_3b:zero_shot	baseline	0.8005	nan	1	0	0	0	0